

The paper investigates a model based on the Assembly Calculus framework, where two key modifications are introduced:

- E%-winners-take-all, where the activation of a neuron depends on its relative input compared to that of the highest input neuron, as opposed to absolute rank
- Inhibitory connections (original only excitatory)

The authors compare the behaviour during assembly formation of the model to the original version, showing both increased stability of assembly formation and orthogonalization of overlapping sensory input in the memory area.

Major overall concerns

The main contribution of the paper remains unclear to me. If I had to come up with a sentence of the style “Adding A and B to the original Assembly Calculus benefits assembly formation via mechanisms X and Y” I would have a hard time identifying X and Y.

Computational papers (usually) fall short on some aspects: analytical results, in-depth explanation of the mechanisms involved in the results, connection to experimental data... But I feel like in its current form the paper does not excel in any of those. The current spirit is probably closest to explaining how the biological mechanism of E%-WTA actually can benefit assembly formation, and that is the main direction I would take to improve the paper.

In this sense, as it stands the manuscript is highly descriptive. Many of the current results could be included as a robustness measure in the supplementary material, with the authors instead using the main text and figures to explain the “why”.

Additionally, there are probably some low-hanging fruit in terms of analytics. Could the authors replicate some of the original derivations with this activation function and show that they match simulations? This might actually also help understand the mechanisms that make this activation function beneficial.

The authors mention how this mechanism is connected to gamma oscillations and power-law distributions. A figure showing this link, some experimental data, and how the model reproduces (or simply has embedded into it) that same data, and how the original AC model does not include it could also be nice.

To be completely transparent, while I am recommending major revisions in this round, I find it unlikely to recommend acceptance if the revised version of the manuscript does not do a good job at mechanistically explaining how the model works, differs (in the process / dynamics, not definition) from the original, and how that benefits assembly formation. I would also be looking to see some effort in either improving the theory behind it or the connection with experiments. I understand the proposed revisions are rather large and maybe the authors would rather take the time to re-submit in the future.

Feedback and clarifying questions

Introduction

I1 - The first paragraph [P2L35-P2L55] perhaps expands too much on the basics of all the background and motivation for the original Assembly Calculus. This is the starting point now so briefly stating what it is and a summary of the original motivation would probably be enough.

I2 - Could the authors explain what they mean when they say that k-WTA dynamics “would not be a concern” if the probability [...] followed a normal distribution”?

I3 - The last paragraph could explain more in detail the results and again aim at describing the findings [instead of “we investigate several properties” say “we find that feed-forward inhibition does X, yielding better pattern separation (for example)]

Minor

I4 - Ref. [25] shows “Anonymous Authors” but probably refers to Meshulam & Bialek (2025)

Methods

M1 - The authors make a great effort to explore the parameter space. However, they leave out of the sweep what is probably one of the most important parameters: epsilon. I understand tau and d are physiological parameters and the used epsilon seems reasonable to me. However, what happens when epsilon goes from 0 to 1? What is the range under which the model behaves well and what breaks out of that range? Depending on the results, you might get some connection with experimental evidence for free (e.g. turns out that the ratio across tau and d in different regions always falls under the well-behaved region, even though the two might co-vary).

M2 - I am not sure synaptic density as defined here is the best metric to use to quantify intra-assembly connectivity. If I understand correctly, given that there is multiplicative plasticity, if a synapse is initialized to zero it will stay 0, and likewise a non-zero synapse cannot go to zero. Seems odd to me to use a metric that is defined based on the number of connections. Maybe I am missing something but if this is the case I would suggest using a metric that takes into account the actual connectivity weight (could be as average sum of incoming connections from other assembly neurons divided by average sum of incoming connections from non-assembly neurons, see similar measures for synaptic assemblies experimentally (Lee et al, *Nature Neuroscience*, 2023)).

M3 - There is a modelling choice of changing the connection probability with respect to the original model (even $p_s \cdot (1 - p_i) = 0.4$, bigger than the original p . At the end of the results we find out why: it needs that to work! It is not necessarily a problem given that additional inhibition has been introduced, but I would like to see in a revised version the mechanism that makes this the case and how it compares to robustness in the original model with p . If both have a sweet spot it is fine, each is competing fairly. If the presented model only wins

for one value of p_s and the other one wins for the rest of values, then the claim “this model is more stable” would be less defensible. In any case this is not an ML conference, if we understand why, given these changes, the probability of connection interacts with assembly formation in a particular manner that is still cool!

Minor

M4 - [P4L23] “In your model” → “In our model”

M5 - I would remove Figure 1b as multiple memory areas are not explored in the paper

M6 - This figure needs better resolution (Most do actually). Also labels should be much bigger, very hard to read

M7 - In 1f), you should also colour the ones active red, ideally specifying what K is. Maybe it would be more useful to understand the difference between the mechanisms to show two input distributions and how changing the highest input changes which ones are active in one case (e.g. 2 in one, 4 in the other), while using a top-2 only two are selected in both cases).

M8 - What is the asterisk in w_{inh} in Table 1?

Results

R1 - Could the authors include in the methods distinct equations for feed-forward and recurrent synaptic initialization? (if I understand correctly only feed-forward contains inhibition). Similarly, please state at the beginning that inhibition is initially absent. If it is always present, please change P12L257-L259 phrasing.

R2 - Would be nice to plot the recurrent connectivity matrix across assemblies (pixel ij is mean connectivity from j to i). (Figure 3).

Minor

R3 - (But very important): Again Figure resolution and label/ticks size

As I consider results need heavy editing, not giving such detailed feedback, but here are some suggestions for improvement. I understand each of the reviewers might come with a full paper proposal of their own. These are only examples of how I would aim at improving in the areas that for me are the weakest at the moment. I would be very happy to propose acceptance if I feel like the revised version of the manuscript has addressed them even though it has been done in a completely different manner to the suggestions here proposed.

1- Finding a title of the form: “Competitive dynamics inspired by gamma oscillations and inhibition benefit assembly formation by doing what we show in Figure 2 and in Figure 4”. (No need to make it the actual title)

Then

2- Using Figure 2 to show mechanistically how learning works in one of the best working examples. Some example questions I think would be interesting to answer:

2.1- How do input statistics and initial random connectivity distribute neural input at the beginning? (Simulations + ideally closed form solution of the initial state if easy to get). Given this input distribution, what K would lead to the same average number of neural assemblies formed?

2.2 - How do synaptic input statistics evolve with time, and how does the associated top-K [the one that at each time interval would make it equivalent to the proposed model] distribution change with time? This is starting to pinpoint how the model differs from the original: If the equivalent K remains approximately the same then this means that the original AC could behave the same if K is chosen right. How does the mean of intra-assembly and inter-assembly synapses evolve with time?

2.3- Now showing a failure case of failure in the original AC model where the proposed model succeeds. What failed in the original model and how did the proposed activation or inhibition mechanism prevent the manuscript model from failing?

The idea is that after looking at Figure 2, the reader understands intuitively how the model works, how that mechanism is different from the original, and how that difference helps learning.

3. Then comes Figure 3. I would personally mix the original manuscript Figure 2 and previous Table 2 into a single revised manuscript Figure 3 saying: "Our model is generally better using these metrics across these parameters". Showing in the figure the parameter(s) that really reflects better stability / convergence and anecdotal magnitudes that are more descriptive can go in a Supplementary Figure.

4. Now: "it can actually do very cool stuff like learning under stressful conditions such as overlapping patterns." Again I would first show the inner workings of one example simulation to explain the mechanism. Then showing the statistics showing when the proposed model is better, not only in that particular example. This could be Figure 4 and the previous Figure 4 can go as a supplementary. I would however again extend it to explain why the model depends on p_s in this manner, and how the original AC behaves with p_s .

5. Figure 5 could show either some analytical results matching simulations or connection with experimental data would make would make the paper more rounded